



Major Task (Part 2)

PHM112: Mathematics (2)

Course Coordinator: Dr. Makram Roshdy

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Deadline: Week

Please, Solve each problem in its assigned place ONLY (the empty space below it)

A. ITERATED INTEGRALS

i) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \int_{-1}^5 \cos y \, dx \, dy$

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos(y) \times \Big|_{-1}^5 = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 5 \cos(y) + \cos(y) \, dy = 6 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos(y) \, dy$$

$$= 6 \left[\sin\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{6}\right) \right] = 3$$



ii) Find the volume of the solid that lies under the plane $3x + 2y + z = 12$ and above the rectangle $R = \{(x, y) \mid 0 \leq x \leq 1, -2 \leq y \leq 3\}$.

$$z = 12 - 3x - 2y$$

$$\int_{-2}^3 \int_0^1 (12 - 3x - 2y) \, dx \, dy$$

$$\int_{-2}^3 \left(12x - \frac{3}{2}x^2 - 2yx \right) \Big|_0^1 \, dy$$

$$\int_{-2}^3 \left(12 - \frac{3}{2} - 2y \right) \, dy$$

$$12y - \frac{3}{2}y - y^2 \Big|_{-2}^3$$

$$\left[12(3) - \frac{3}{2}(3) - (3)^2 \right] - \left[12(-2) - \frac{3}{2}(-2) - (-2)^2 \right]$$

$$\frac{45}{2} + 25 = 47.5$$

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- iii) Find the volume of the solid in the first octant bounded by the cylinder $z = 16 - x^2$ and the plane $y = 5$.

$$16 - x^2 = 0$$

$x = \pm 4 \Rightarrow$ for first octant $-4 \Rightarrow$ use $(0, 4)$

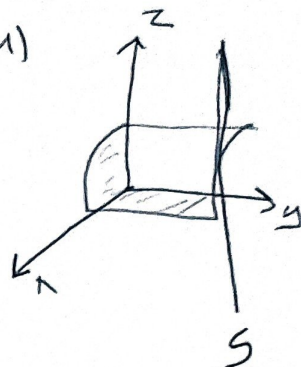
$$y \Rightarrow (0, 5)$$

$$z \Rightarrow (0, 16 - x^2)$$

$$\int_0^5 \int_0^4 \int_0^{16-x^2} dz dx dy = \int_0^5 \int_0^4 (16 - x^2) dx dy$$

$$\int_0^5 \left(16x - \frac{1}{3}x^3 \right) \Big|_0^4 dy = \int_0^5 \left(64 - \frac{64}{3} \right) dy$$

$$\frac{128}{3} y \Big|_0^5 = 213.33$$



B. DOUBLE INTEGRALS OVER GENERAL REGIONS

- i) Find the volume of the solid enclosed by the paraboloid $z = x^2 + 3y^2$ and the planes $x = 0$, $y = 1$, $y = x$, $z = 0$.

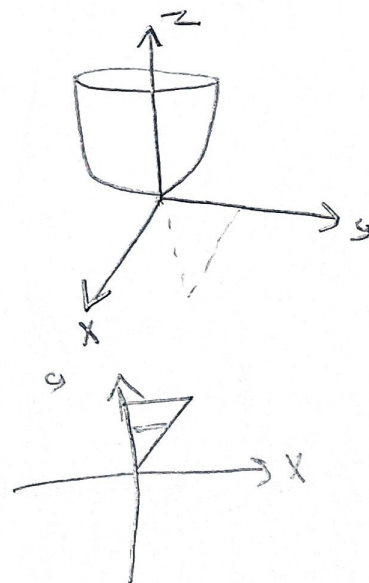
$$\int_0^1 \int_0^y (x^2 + 3y^2) dx dy$$

$$\int_0^1 \left(\frac{1}{3}x^3 + 3y^2x \right) \Big|_0^y dy$$

$$\int_0^1 \left(\frac{1}{12}y^3 + 3y^3 \right) dy$$

$$\left(\frac{1}{48}y^4 + \frac{3}{4}y^4 \right) \Big|_0^1$$

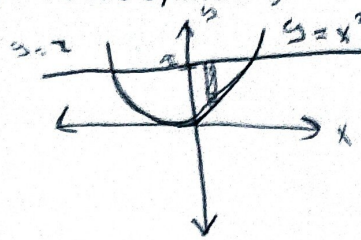
$$= \frac{5}{6}$$



- ii) Find the volume of the solid enclosed by the parabolic cylinder $y = x^2$ and the planes $z = 3y$, $z = 2 + y$.

$$3y = 2 + y$$

$$x = \pm 1$$



$$\int_{-1}^1 \int_0^1 (3y - 2 - y) dy dx$$

$$\int_{-1}^1 \left(\frac{3}{2} y^2 - 2y - \frac{1}{2} y^2 \right) \Big|_0^1 dx$$

$$\int_{-1}^1 \left(\frac{3}{2} - 2 - \frac{1}{2} \right) dx = \int_{-1}^1 -1 dx = -1 - 1 = -2$$

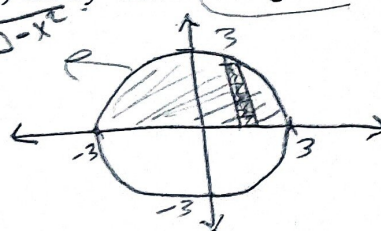
- iii) Sketch the region of integration: $\int_0^3 \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} f(x, y) dx dy$ hence change the order.

$$x = \sqrt{9-y^2}$$

$$x^2 + y^2 = 9$$

$$y = 0 \quad y = 3$$

$$y = \sqrt{9-x^2}$$



$$\int_{-3}^3 \int_0^{\sqrt{9-x^2}} f(x, y) dy dx$$

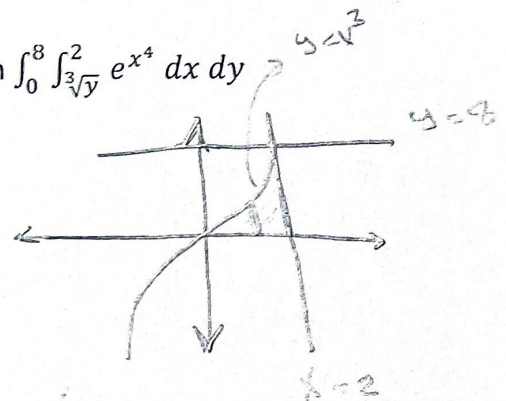
- iv) Evaluate the integral by reversing the order of integration $\int_0^8 \int_{\sqrt[3]{y}}^2 e^{x^4} dx dy$

$$\int_0^2 \int_0^{x^3} e^{x^4} dy dx$$

$$\frac{1}{4} \int_0^2 4x^3 e^{x^4} dx$$

$$\frac{1}{4} [e^{x^4}]_0^2$$

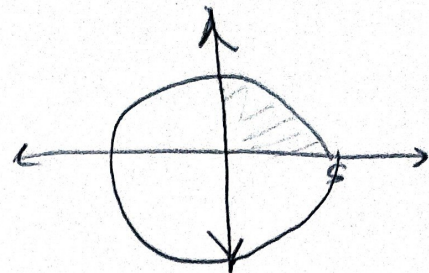
$$\frac{1}{4} [e^{16} - 1]$$



C. DOUBLE INTEGRALS IN POLAR COORDINATES

- i) Evaluate $\iint_R ye^x dA$ by changing to polar coordinates where R is the region in the first quadrant enclosed by the circle $x^2 + y^2 = 25$.

$$\int_0^{\frac{\pi}{2}} \int_0^5 r \sin(\phi) e^{r \cos(\phi)} dr d\phi$$



- ii) Use Polar coordinates to find the volume of the given solid.

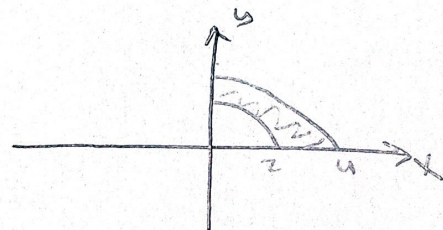
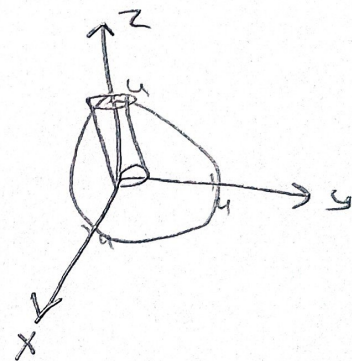
1. Inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$.

$$\int_0^{\frac{\pi}{2}} \int_2^4 2\sqrt{16-r^2} r dr d\phi$$

$$\int_0^{\frac{\pi}{2}} \frac{1}{2} \int_2^4 4(16-r^2)^{\frac{1}{2}} r dr d\phi$$

$$\frac{\pi}{2} \times 2 \times \frac{1}{2} \int_2^4 -2(16-r^2)^{\frac{1}{2}} r dr$$

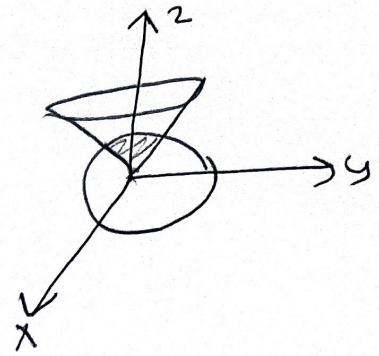
$$= 13.86 W$$



2. Above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$.

$$z = 1$$

$$z = \sqrt{1-r^2}$$



$$2\pi \frac{\sqrt{2}}{2}$$

$$\int_0^{\sqrt{2}/2} \int_0^{2\pi} (\sqrt{1-r^2} - r) r dr d\phi$$

$$2\pi \int_0^{\sqrt{2}/2} r (\sqrt{1-r^2} - r) dr$$

$$2\pi \left[-\frac{1}{2} \times \frac{2}{3} (1-r^2)^{3/2} - \frac{r^2}{2} \right]_0^{\sqrt{2}/2}$$

$$= \frac{2\pi}{3} \left(1 - \frac{1}{\sqrt{2}} \right)$$

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iii) Given that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, evaluate

$$1. \int_0^{\infty} x^2 e^{-x^2} dx$$

$$2. \int_0^{\infty} \sqrt{x} e^{-x} dx$$

$$I_1 = \int_{-\infty}^{\infty} e^{-x^2} dx, I_2 = \int_{-\infty}^{\infty} e^{-y^2} dy$$

$$(I_1 I_2)^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2 - y^2} dx dy = \pi$$

$$= \frac{1}{2} [\infty \cdot 0] + \frac{1}{2} \int_0^{\infty} e^{-x^2} dx$$

$$\lim_{x \rightarrow 0} \frac{x}{e^{x^2}} = \frac{1}{\infty} = 0$$

D. Triple Integral

1. Use a triple integral to find the volume of the solid enclosed by the paraboloid $x = y^2 + z^2$ and the plane $x = 16$.

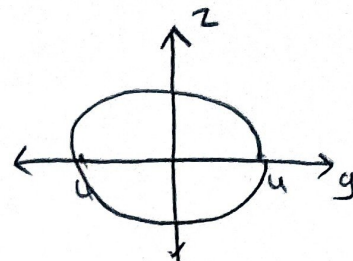
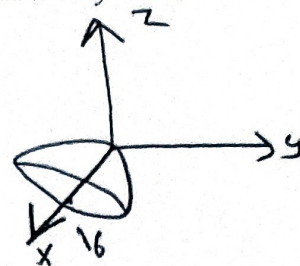
$$\iiint dx \, dy \, dz$$

$$\int_0^{2\pi} \int_0^4 \int_{y^2+z^2}^{16} dx \, dy \, dz$$

$$\int_0^{2\pi} \int_0^4 \int_{y^2+z^2}^{16} 1 \, dx \, dy \, dz = \int_0^{2\pi} \int_0^4 (16 - y^2 - z^2) \, dy \, dz$$

$$\int_0^{2\pi} \left[\frac{16}{2} y^2 - \frac{1}{4} y^4 \right]_0^4 \, dz$$

$$\int_0^{2\pi} (128 - 64) \, dz = \int_0^{2\pi} 64 \, dz = 128\pi$$



2. Sketch the solid whose volume is given by the iterated integral.

$$\int_0^1 \int_0^{1-x} \int_0^{2-2z} dy \, dz \, dx$$

$$y = 2 - 2z$$

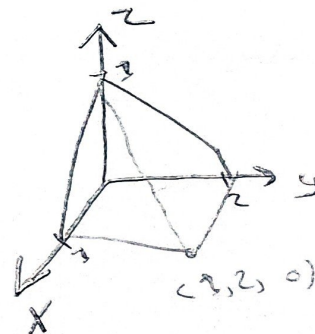
$$y = 2z$$

$$z = 1 - x$$

$$z = 2x$$

$$x = 1$$

$$x = 2z$$



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3. Sketch the solid whose volume is given by the integral and evaluate the integral.

$$z = 9 - (x^2 + y^2)$$

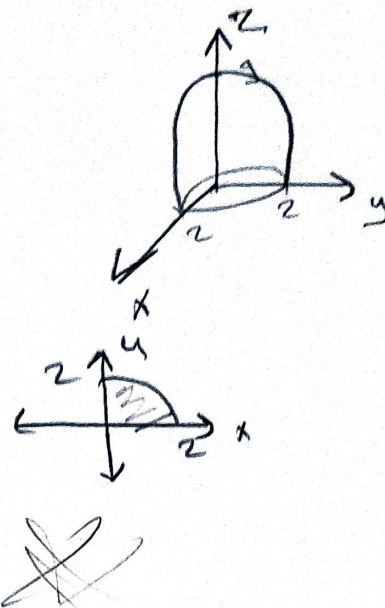
$$z = 9 - x^2 - y^2$$

$$\int_0^{\pi/2} \int_0^2 \int_0^{9-r^2} r \, dz \, dr \, d\theta$$

$$\int_0^{\pi/2} \int_0^2 r(9-r^2) \, dr \, d\theta$$

$$\int_0^{\pi/2} \left[\frac{9}{2} r^2 - \frac{1}{4} r^4 \right]_0^2 d\theta$$

$$= \int_0^{\pi/2} 14 \, d\theta = \frac{\pi}{2} \times 14 = 7\pi$$



4. Use cylindrical coordinates to

A) evaluate $\iiint_E e^z \, dV$, where E is enclosed by the paraboloid $z = 1 + x^2 + y^2$, the cylinder $x^2 + y^2 = 5$ and the xy -plane.

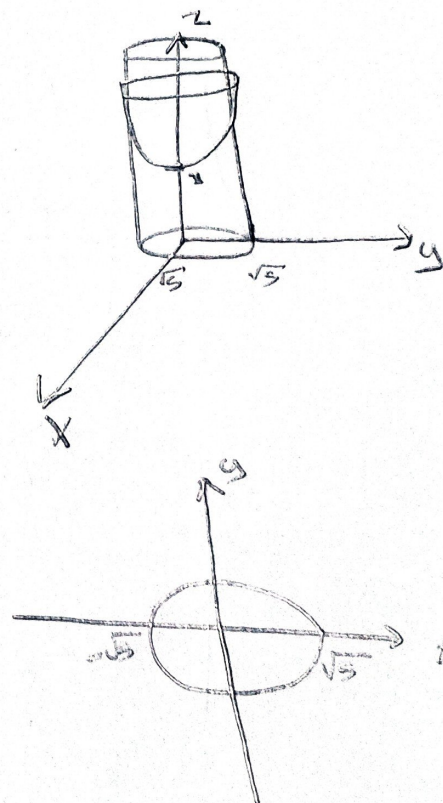
$$\int_0^{2\pi} \int_0^{\sqrt{5}} \int_0^{1+r^2} e^z \, r \, dz \, dr \, d\theta$$

$$\int_0^{2\pi} \int_0^{\sqrt{5}} (e^{1+r^2} - 1) \, r \, dr \, d\theta$$

$$\int_0^{2\pi} \left[\frac{1}{2} e^{1+r^2} + \frac{1}{4} e^{1+r^2} \right]_0^{\sqrt{5}} d\theta$$

$$\int_0^{2\pi} 8.75 \, d\theta$$

$$= 17.5\pi$$



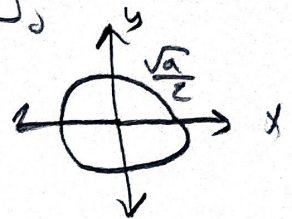
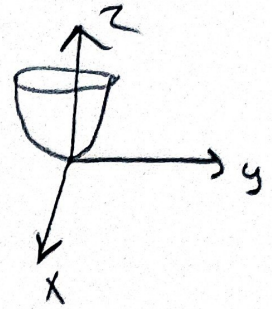
B) Find the mass of the solid S bounded by the paraboloid $z = 4x^2 + 4y^2$ and the $z = a$ ($a > 0$) if S has constant density K .

$$m = \iiint_S \rho \, dV$$

$$= \int_0^{2\pi} \int_0^{\frac{\sqrt{a}}{2}} \int_{4x^2+4y^2}^a dz \, r \, dr \, d\phi$$

$$2\pi K \int_0^{\frac{\sqrt{a}}{2}} (a - 4r^2) \times r \, dr = 2\pi K \left[\frac{ar^2}{2} - \frac{4r^4}{4} \right]_0^{\frac{\sqrt{a}}{2}}$$

$$m = \frac{a^2 \pi K}{8}$$



5. Sketch the solid whose volume is given by the integral and evaluate the integral.

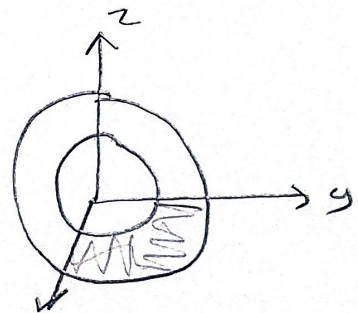
$$\int_0^{2\pi} \int_{\pi/2}^{\pi} \int_1^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$\int_0^{2\pi} \int_{\pi/2}^{\pi} \frac{\rho^3}{3} \sin \phi \, d\phi \, d\theta$$

$$\int_0^{2\pi} \int_{\pi/2}^{\pi} \frac{7}{3} \sin \phi \, d\phi \, d\theta$$

$$\int_0^{2\pi} \frac{7}{3} \cos \phi \Big|_{\pi/2}^{\pi} d\theta$$

$$\int_0^{2\pi} \frac{7}{3} d\theta = \frac{14}{3} \pi$$



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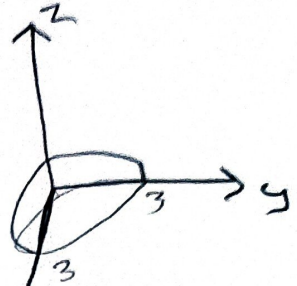
6. Use spherical coordinates to evaluate $\iiint_E e^{\sqrt{x^2+y^2+z^2}} dV$, where E is enclosed by the sphere $x^2 + y^2 + z^2 = 9$ in the 1st octant.

$$\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^3 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$\rho^2 e^{\rho} - 2\rho e^{\rho} + \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^3 2e^{\rho} \sin \phi \, d\rho \, d\phi \, d\theta$$

$$3e^3 + \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} 2(e^3 - 1) \sin \phi \, d\phi \, d\theta$$

$$= 3e^3 + e^3\pi - \pi$$



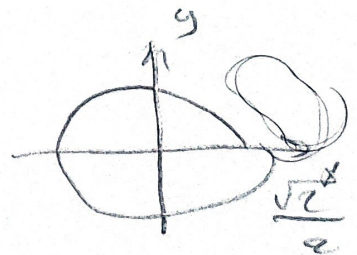
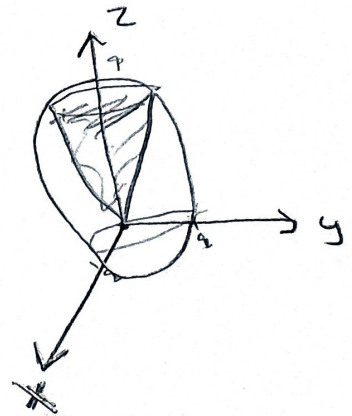
7. Use cylindrical or spherical coordinates (whichever seems more appropriate) to find the volume of solid E that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$.

$$\int_0^{2\pi} \int_0^{\frac{\sqrt{2}}{2}} \int_{\sqrt{1-r^2}}^{\sqrt{1-r^2}} dz \, r \, dr \, d\theta$$

$$\int_0^{2\pi} \int_0^{\frac{\sqrt{2}}{2}} (1 - (1-r^2)^{\frac{1}{2}} - r^2) r \, dr \, d\theta$$

$$\int_0^{2\pi} d\theta \times \int_0^{\frac{\sqrt{2}}{2}} -2(1-r^2)^{\frac{1}{2}} - r^2$$

$$= 3\pi \left[-\frac{1}{3\sqrt{2}} - \frac{\sqrt{2} + 1}{12} \right]$$



E. Line Integral and Green Theorem

1. Evaluate the following line integral, where C a given curve.

$$\int_C x^2 y \sqrt{z} ds, \quad C: x = t^3, y = t, z = t^2, \quad 0 \leq t \leq 1$$

$$\int_0^1 t^0 \cdot t \cdot 2t dt$$

$$\int_0^1 2t^2 dt = \frac{1}{5}$$

2. Find the arc length of the curve $y^2 = x^3$ between the points (1,1) and (4,8).

$$2yy' = 3x^2$$

$$y = x^{\frac{3}{2}}$$

$$y' = \frac{3x^2}{2y}$$

$$\int_1^4 \sqrt{1 + \left(\frac{3x^2}{2y}\right)^2} dx = \int_1^4 \left(1 + \frac{9}{4}x\right)^{\frac{1}{2}} dx$$

$$= 7.63$$



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3. Show that the line integral $\int_{(1,0)}^{(2,1)} (2xy - y^4 + 3)dx + (x^2 - 4xy^3)dy$ is path independent hence evaluate it.

$$\begin{aligned} 2xy - y^4 + 3 & \quad x^2 - 4xy^3 \\ \frac{d}{dy} (2xy - y^4 + 3) & \quad \frac{d}{dx} (x^2 - 4xy^3) \\ = 2x - 4y^3 & \quad = 2x - 4y^3 \end{aligned}$$

$$\begin{aligned} y=0 & \Rightarrow x=1 \Rightarrow y=x-1 & x=y+1 \\ x=1 & & x'=1 \end{aligned}$$

$$\begin{aligned} \int_0^1 2(2y+1)y - y^4 + 3(y+1)^2 - 4(y+1)y^3 dy \\ = \underline{\underline{5}} \end{aligned}$$

4. Use Green's Theorem to evaluate the following line integral along the given positively oriented curve.

$\int_C y^3 dx - x^3 dy$, C is the circle $x^2 + y^2 = 4$